

Algebra 1
Unit 8
Lesson 2 Homework

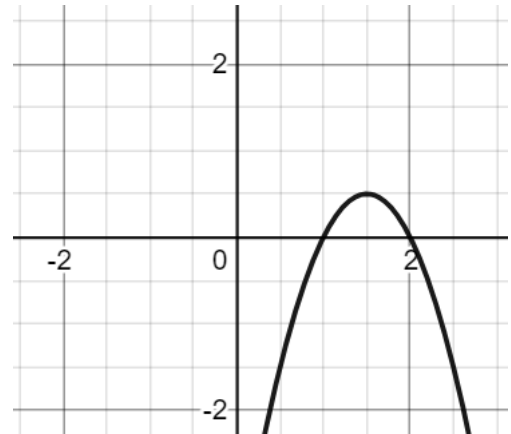
Name **Answer Key** _____
Date _____
Period _____

1. As the x - values of the parabola get infinitely *larger*, the y -values of the parabola approach

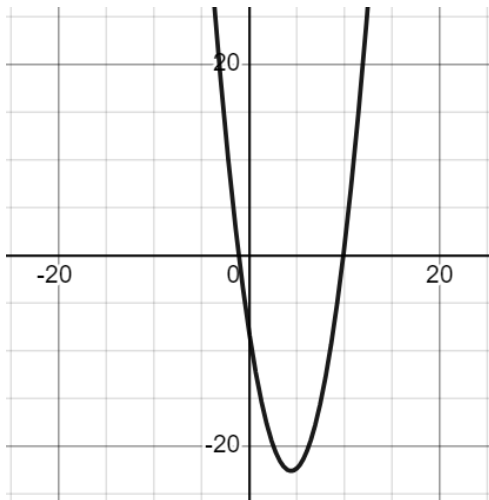
☐ ∞
☒ $-\infty$

2. As the x -values of the parabola get infinitely *smaller*, the y -values of the parabola approach

☐ ∞
☒ $-\infty$



Use the graph of the quadratic equation for questions 3 and 4.



3. Use words or symbols to describe the domain of the quadratic function.

All real #'s

4. Use words or symbols to describe the range of the quadratic function.

$y \geq -22$

5. Determine the vertex and y -intercept of a quadratic using the equation $y = -5(x - 1)^2 + 3$

$y = -5(0 - 1)^2 + 3$
 $y = -2$

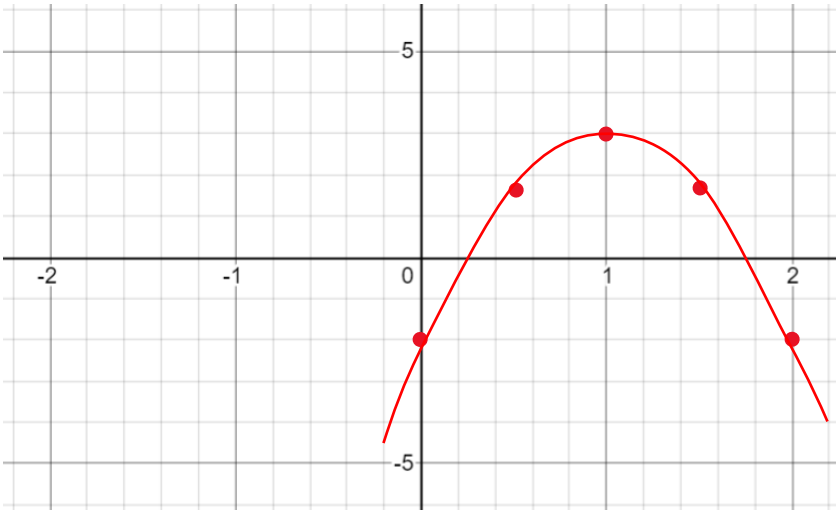
vertex = (1,3)
 y -int = (0, -2)

6. Use the table of values to determine the vertex of the parabola.

vertex = (1,3)

x	y
0.1	-1.05
0.2254	0
0.5	1.75
1	3
1.5	1.75
1.7746	0
2	-2

7. Use the key features you determined in question 5 and 6 to sketch the graph of the quadratic function.



8. Identify key features of the quadratic $y = -(x - 3)^2 + 4$

x -intercept	(1,0)	vertex	(3,4)	domain	All Real #’s	left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \underline{-\infty}$
x -intercept	(5,0)	increasing	$(-\infty, 3)$	range	$y \leq 4$	right end behavior	$x \rightarrow \infty,$ $y \rightarrow \underline{-\infty}$
y -intercept	(0,−5)	decreasing	$(3, -\infty)$	opens	Down		

9. Create a table of values for the quadratic function in question 8.

x	−2	−1	0	1	2	3
y	−21	−12	−5	0	3	4

10. Use the key features and table of values from 8 and 9 to graph the quadratic.

